

Report No.:

Test Time: 2024/10/15 11:13:32

LUMINAIRE PHOTOMETRIC TEST REPORT

Luminaire Property

Manufacturer: 0

Luminaire Type: AL9331 4000K 80RA

Lamp Type:

Luminous Width: 0 mm

Luminous Length: 0 mm

Luminous Height: 0 mm

Voltage(V): 219.50 V

Current(A): 0.0360 A

Power(W): 7.30 W

Power Factor: 0.9130

Lamp Luminous Flux: 493.73

Photometric Results

CIE Class: Direct

Luminaire Luminous Flux: 493.7257 lm

Downward Ratio: 100.00%

Luminaire Efficacy Rating(LER): 67.63 lm/W

Beam Angle(50%) C0-180,C90-270: 86.4,86.5

Field Angle(10%) C0-180,C90-270: 132.1,131.9

S/MH C0-180,C90-270: 1.155,1.155

Max. Intensity: 256.76 cd

Max.Intensity Angle: C=180 Gamma=3

Upward Ratio: 0.00%

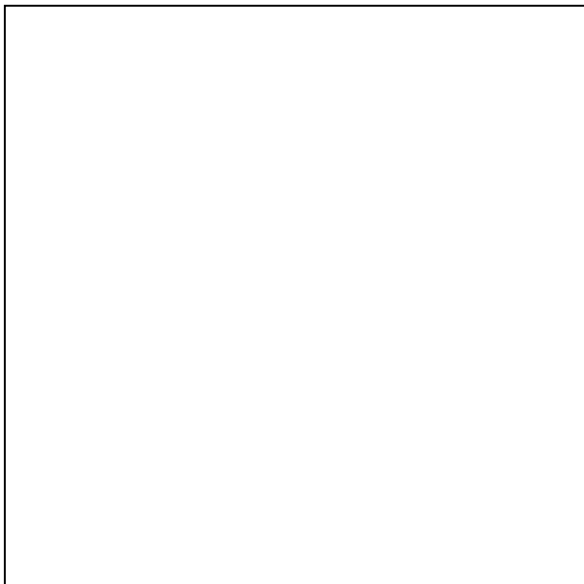
Efficacy: 100.00%

Center Intensity: 256.72 cd

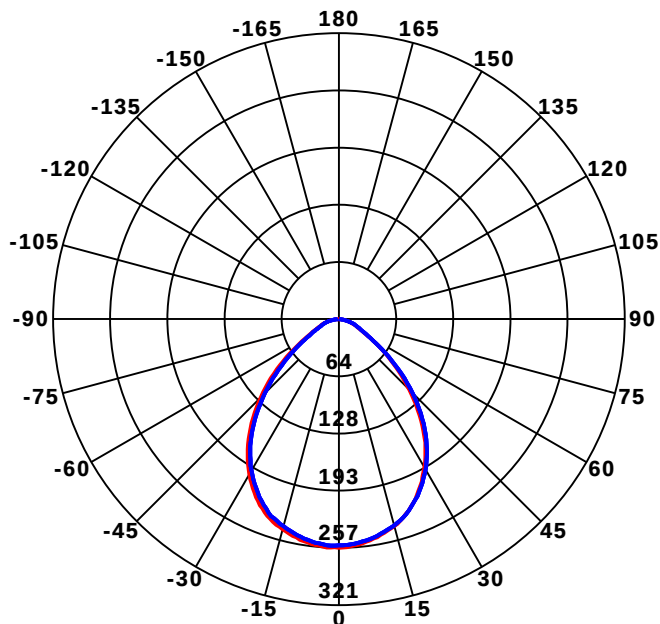
Effective Luminous Flux: 465.56 lm

ERP(φ use)(120deg): 445.81 lm

Picture of Luminaire



Luminous Intensity Distribution Curve



Beam Angle(50%) C0-180,C90-270: 86.4 deg,86.5 deg

C0-180 — C90-270 —

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

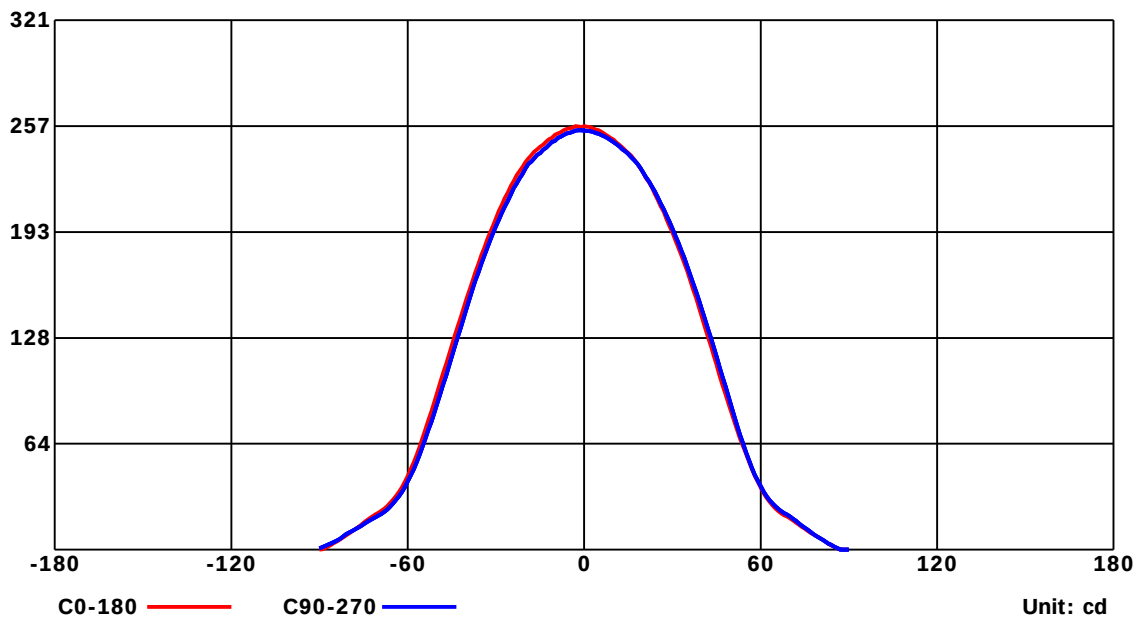
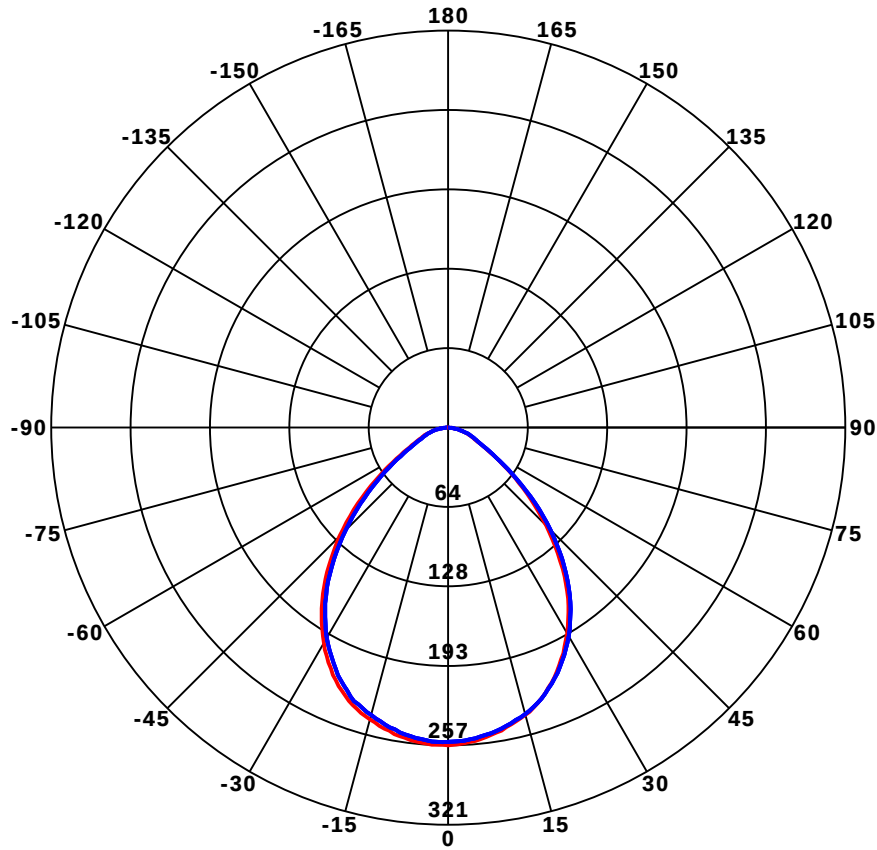
Test Distance: 6.25 m

Remarks:

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Luminous Intensity Distribution Curve



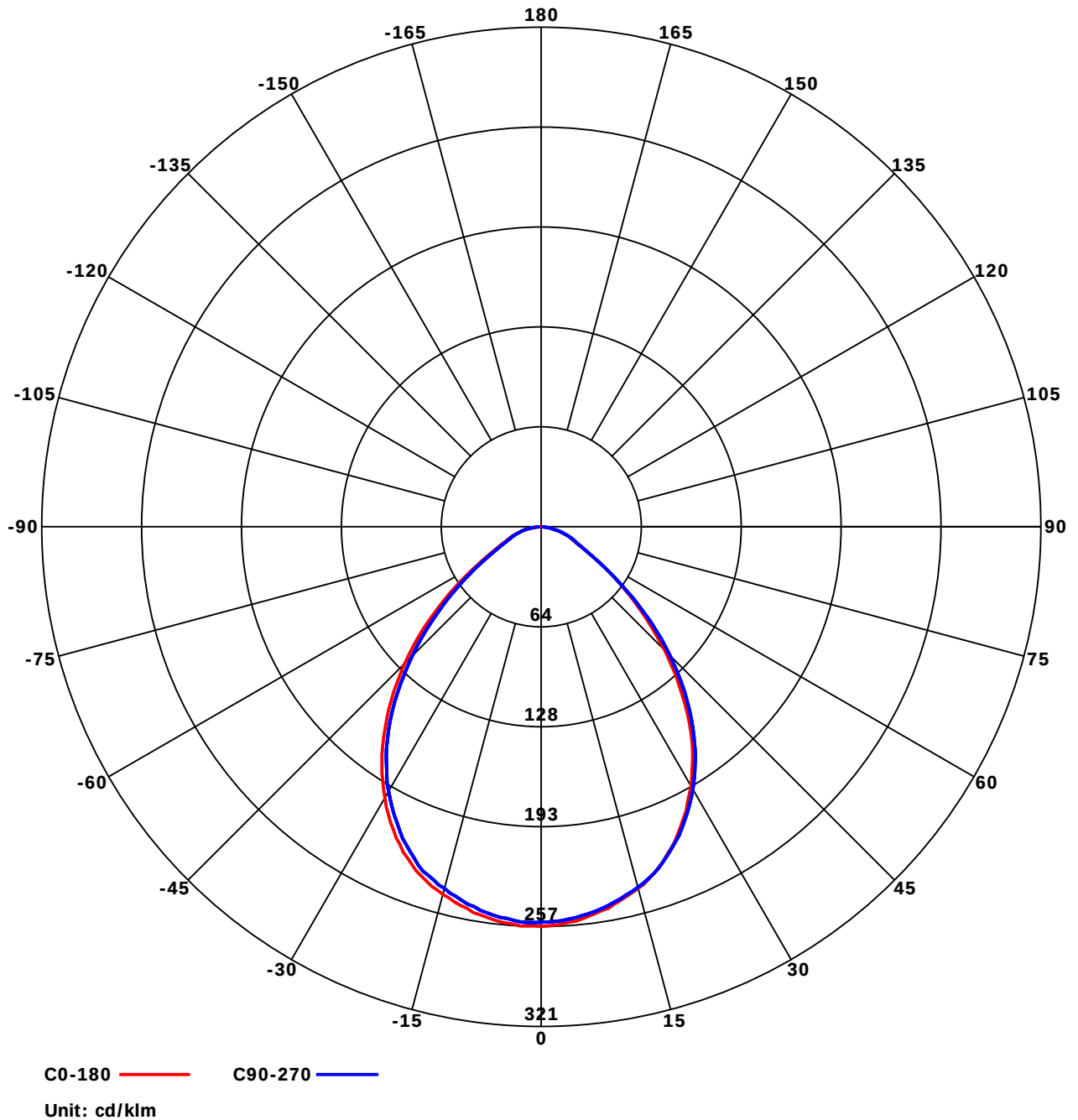
C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

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Luminous Intensity Distribution Curve



C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

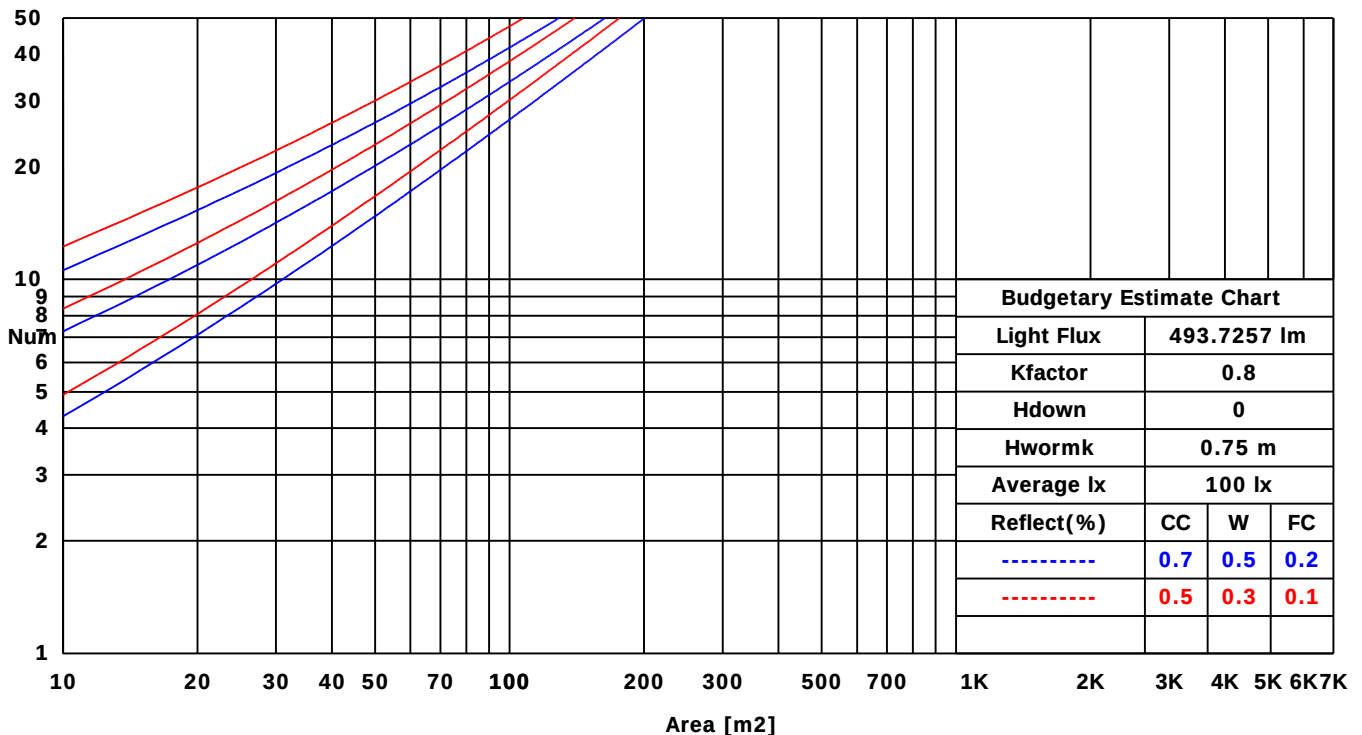
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UTILIZATION FACTORS TABLE

RHOCC	80			70			50			30			10			0
RHOW	50	30	10	50	30	10	50	30	10	50	30	10	50	30	10	0
RCR	COEFFICIENTS OF UTILIZATION FOR RHOFC=20															
0	1.18	1.18	1.18	1.15	1.15	1.15	1.10	1.10	1.10	1.06	1.06	1.06	1.01	1.01	1.01	0.99
1	1.08	1.07	1.06	1.07	1.05	1.04	1.03	1.01	1.00	0.98	0.96	0.94	0.91	0.89	0.88	0.82
2	0.96	0.94	0.92	0.95	0.92	0.90	0.92	0.89	0.87	0.88	0.85	0.82	0.83	0.80	0.77	0.72
3	0.85	0.82	0.81	0.84	0.81	0.79	0.82	0.79	0.76	0.79	0.75	0.72	0.75	0.71	0.67	0.63
4	0.75	0.73	0.71	0.75	0.72	0.70	0.74	0.70	0.67	0.72	0.67	0.63	0.69	0.64	0.60	0.56
5	0.67	0.65	0.63	0.67	0.64	0.62	0.67	0.62	0.59	0.65	0.60	0.56	0.63	0.58	0.53	0.49
6	0.60	0.58	0.57	0.61	0.58	0.55	0.61	0.56	0.53	0.60	0.54	0.51	0.58	0.52	0.48	0.44
7	0.55	0.52	0.51	0.55	0.52	0.50	0.55	0.51	0.48	0.55	0.49	0.46	0.54	0.48	0.43	0.40
8	0.50	0.48	0.46	0.50	0.47	0.45	0.51	0.46	0.43	0.51	0.45	0.41	0.50	0.44	0.39	0.36
9	0.46	0.43	0.42	0.46	0.43	0.41	0.47	0.42	0.40	0.47	0.41	0.38	0.46	0.40	0.36	0.33
10	0.42	0.40	0.39	0.42	0.40	0.38	0.43	0.39	0.36	0.43	0.38	0.35	0.43	0.37	0.33	0.30

LUMINAIRE BUDGETARY ESTIMATE DIAGRAM



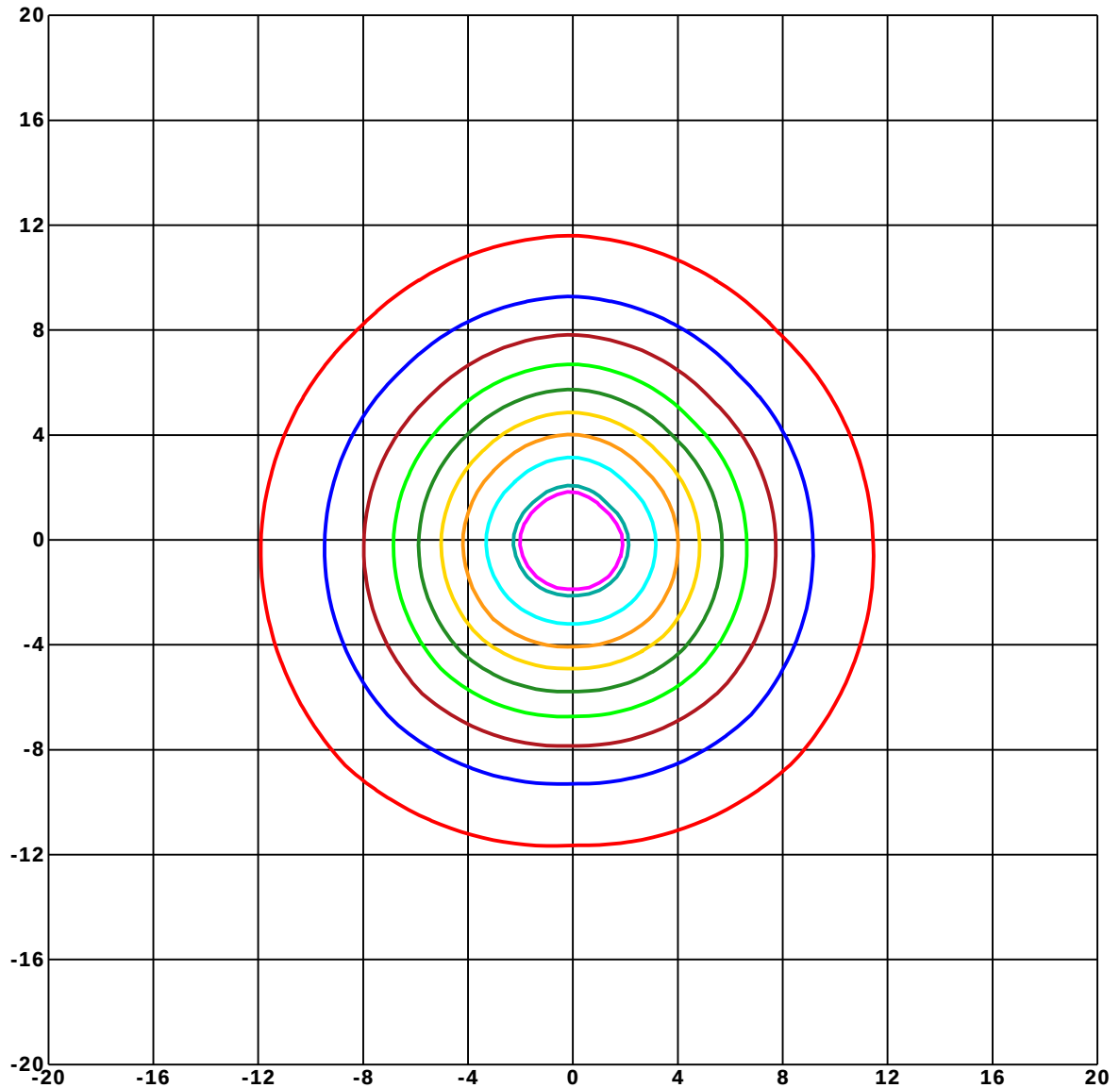
C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

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ISOLX DIAGRAM



Unit of illumination:lx Height of plane:10.0(m)/32.81(ft)

2.362(92%Emax)	1.284(50%Emax)
2.31(90%Emax)	1.027(40%Emax)
2.054(80%Emax)	0.77(30%Emax)
1.797(70%Emax)	0.513(20%Emax)
1.54(60%Emax)	0.257(10%Emax)

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

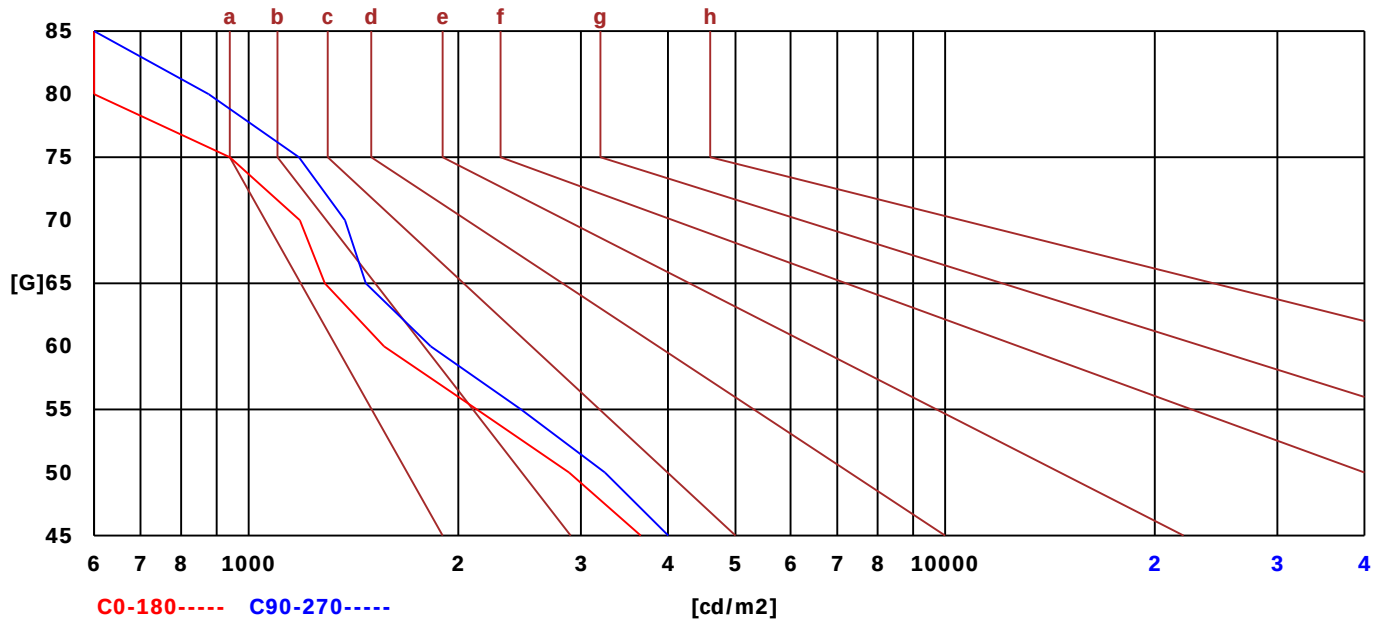
Remarks:

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BRIGHT LIMIT DIAGRAM

C/G	45	50	55	60	65	70	75	80	85
C0	3656	2884	2127	1566	1287	1185	940	562	0
C90	4005	3247	2462	1826	1474	1376	1181	877	168
Dazzle	Quality	Illuminance	---	---	---	---	---	---	---
1.15	A	2000	1000	500	<=300				
1.5	B		2000	1000	500	<=300			
1.85	C			2000	1000	500	<=300		
2.2	D				2000	1000	500	<=300	
2.55	E					2000	1000	500	<=300
---	---	a	b	c	d	e	f	g	h



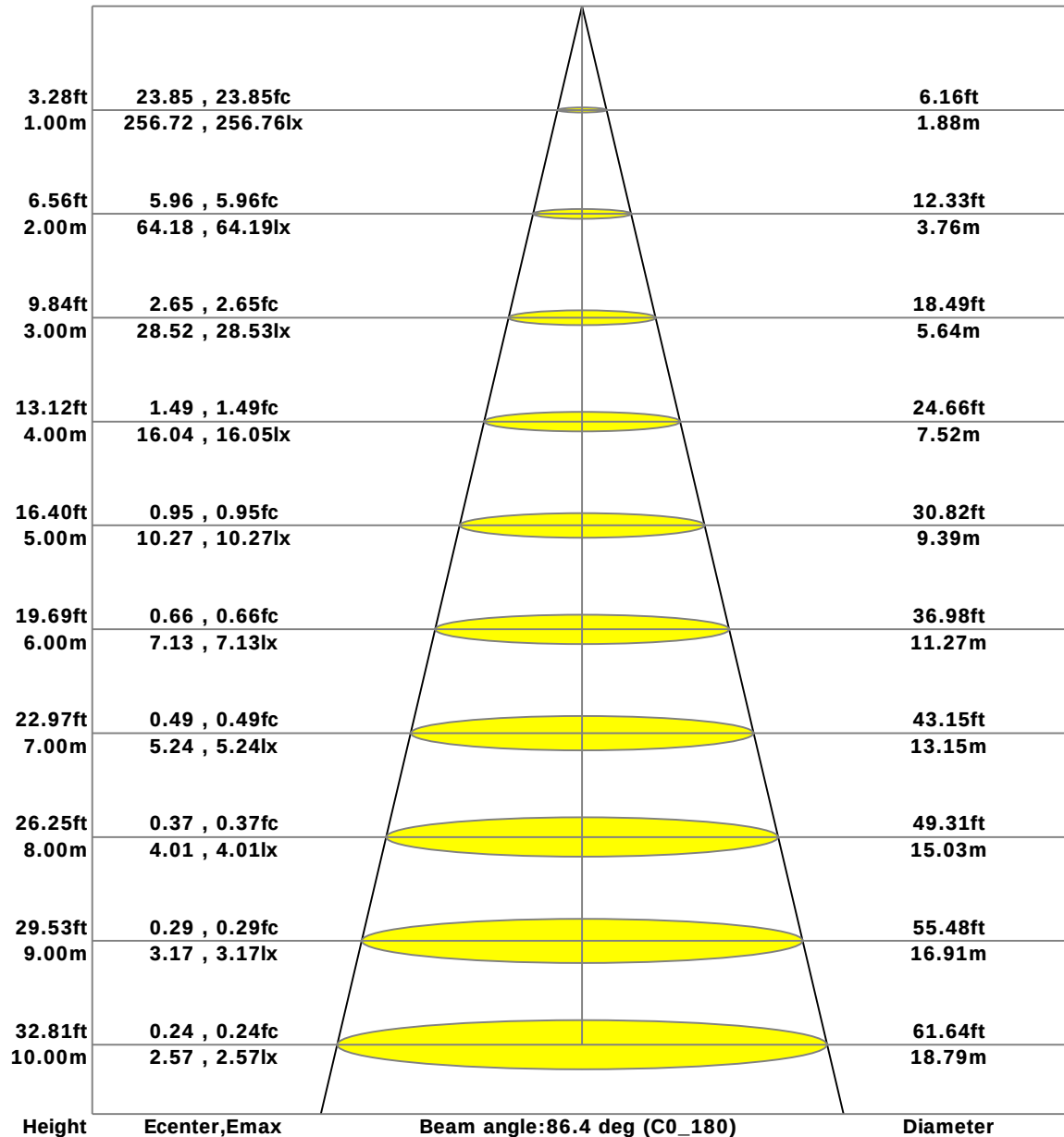
C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

Report No.:

Test Time: 2024/10/15 11:13:32

LX-DISTANCE DIAGRAM



C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:13:32

UGR(Unified Glare Rating) TABLE

RHOCC		70	70	50	50	30	70	70	50	50	30
RHOW		50	30	50	30	30	50	30	50	30	30
RHOF		20	20	20	20	20	20	20	20	20	20
Room Size X Y		Viewed Endwise					Viewed Crosswise				
2H 2H		15.06	15.63	14.99	15.81	16.06	15.15	15.72	15.08	15.90	16.15
3H		14.74	15.26	14.73	15.48	15.78	14.83	15.35	14.82	15.57	15.87
4H		14.55	15.05	14.58	15.29	15.61	14.64	15.14	14.67	15.37	15.70
6H		14.35	14.82	14.42	15.08	15.43	14.44	14.90	14.50	15.16	15.52
8H		14.25	14.70	14.33	14.97	15.34	14.33	14.78	14.42	15.05	15.42
12H		14.14	14.58	14.25	14.86	15.24	14.23	14.66	14.33	14.94	15.32
4H 2H		15.62	16.12	15.65	16.35	16.68	15.76	16.25	15.78	16.49	16.81
3H		15.35	15.78	15.45	16.06	16.45	15.48	15.92	15.59	16.20	16.58
4H		15.17	15.57	15.32	15.87	16.29	15.30	15.70	15.45	16.00	16.42
6H		14.96	15.32	15.16	15.64	16.10	15.09	15.45	15.29	15.77	16.23
8H		14.85	15.18	15.07	15.52	16.00	14.98	15.31	15.20	15.65	16.13
12H		14.74	15.05	14.99	15.40	15.89	14.87	15.18	15.12	15.53	16.02
8H 4H		15.61	16.06	15.69	16.33	16.69	15.74	16.19	15.83	16.46	16.83
6H		15.00	15.28	15.27	15.64	16.16	15.12	15.40	15.40	15.77	16.28
8H		14.89	15.14	15.19	15.51	16.05	15.01	15.26	15.31	15.64	16.18
12H		14.78	15.00	15.11	15.39	15.94	14.90	15.12	15.23	15.51	16.07
12H 4H		15.12	15.42	15.37	15.78	16.27	15.25	15.56	15.50	15.91	16.40
6H		14.92	15.17	15.22	15.54	16.08	15.05	15.30	15.35	15.67	16.21

C Range: 0 - 360 Deg
 C Interval: 45.0 Deg
 Test Speed: High
 Temperature: 6.25
 Operators:
 Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg
 G Interval: 1.0 Deg
 Test System: Huipu CPM-2000B Goniophotometer
 Humidity: 1
 Test Distance: 6.25 m
 Remarks:

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ZONAL LUMINOUS FLUX TABLE-1

[illegible]

C Range: 0 - 360 Deg
C Interval: 45.0 Deg
Test Speed: High
Temperature: 6.25
Operators:
Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg
G Interval: 1.0 Deg
Test System: Huipu CPM-2000B Goniophotometer
Humidity: 1
Test Distance: 6.25 m
Remarks:

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ZONAL LUMINOUS FLUX TABLE-2

γ	C0	C45	C90	C135	C180	C225	C270	C315	γ	φ zonal	φ total	%lum	%lamp
10	248.7	243.1	247.1	246.9	251.6	249.6	247.8	249.2	10	24.0	24.0	4.9	4.9
20	228.8	221.5	228.9	228.4	235.0	235.7	231.2	233.9	20	67.8	91.8	18.6	18.6
30	192.3	183.8	194.7	192.3	201.1	207.5	196.2	202.4	30	98.9	190.7	38.6	38.6
40	140.9	132.6	145.6	142.1	152.4	162.4	146.3	154.6	40	108.0	298.6	60.5	60.5
50	83.2	74.1	87.0	83.5	95.3	104.5	88.5	96.3	50	90.9	389.6	78.9	78.9
60	37.5	31.3	38.4	36.5	44.8	52.8	41.4	46.6	60	56.2	445.8	90.3	90.3
70	18.9	16.2	20.0	18.8	22.3	25.0	20.6	22.7	70	28.6	474.4	96.1	96.1
80	7.3	3.9	7.1	6.1	9.8	13.1	10.4	11.3	80	15.3	489.7	99.2	99.2
90	0.0	0.0	0.0	0.0	0.0	2.8	0.9	1.5	90	4.0	493.7	100.0	100.0

Conical surface Flux(90deg) : 347.8 lm

%lum = 70.4%

%lamp = 70.4%

Conical surface Flux(120deg) : 445.8 lm

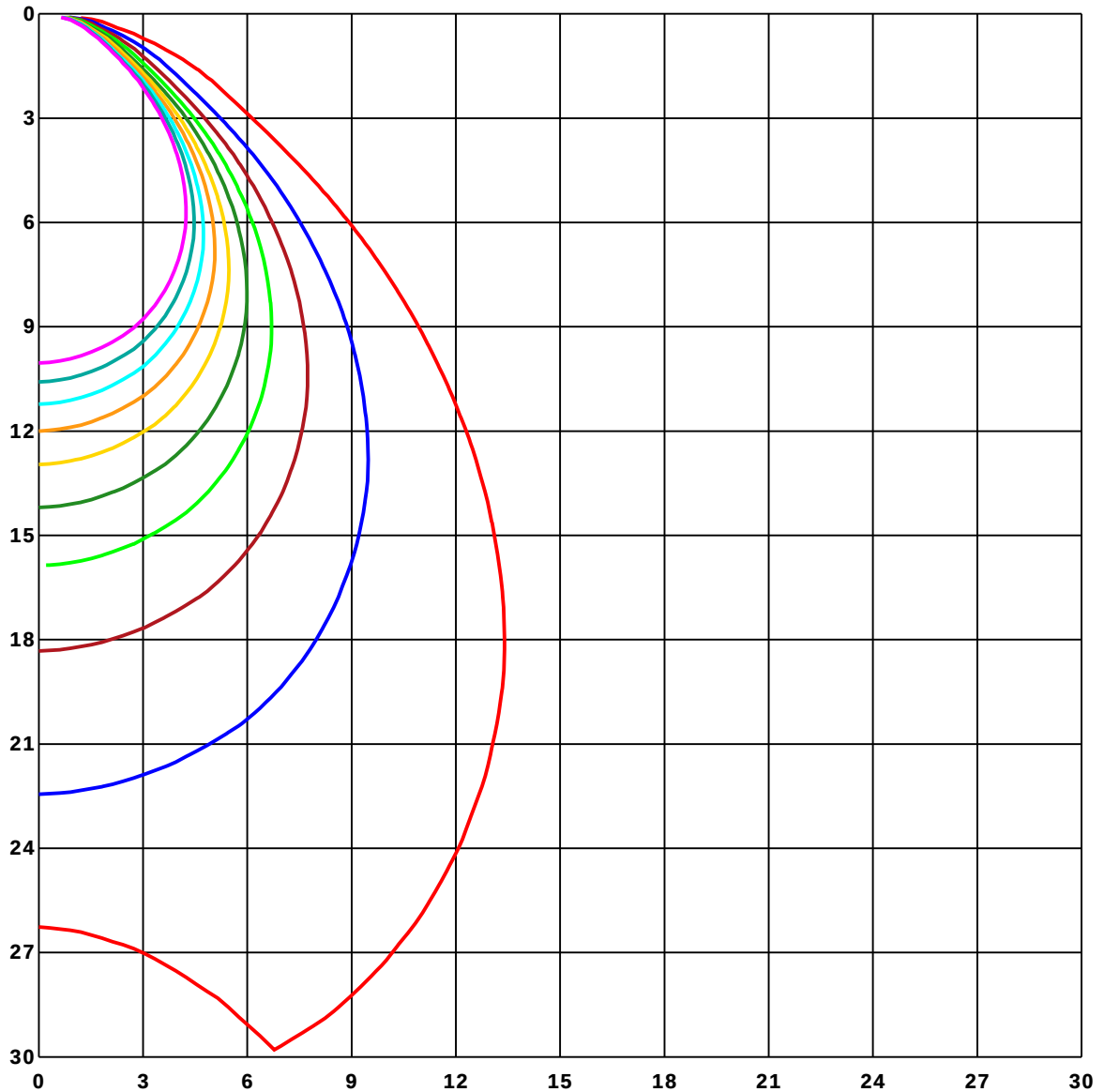
%lum = 90.3%

%lamp = 90.3%

Report No.:

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SPACE ISOLX DIAGRAM



Unit of illumination:lx

2.548
 2.293
 2.039
 1.784
 1.529

1.274
 1.019
 0.764
 0.51
 0.255

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:13:32

LUMINOUS DISTRIBUTION INTENSITY DATA

Gamma/C	C0	C45	C90	C135	C180	C225	C270	C315	C360	
G0	256.72	253.05	254.02	254.84	256.72	253.05	254.02	254.84	256.72	
G1	256.29	252.62	253.91	254.84	256.25	253.13	254.22	255.08	256.29	
G2	256.09	252.03	253.67	254.38	256.56	253.05	254.26	255.16	256.09	
G3	255.39	250.98	253.24	254.3	256.76	253.59	254.14	254.73	255.39	
G4	254.57	250.35	252.54	253.32	255.86	253.09	253.36	254.34	254.57	
G5	254.18	249.38	252.07	252.93	255.39	252.81	252.62	253.83	254.18	
G6	253.16	248.24	251.05	251.68	254.96	252.27	252.3	253.2	253.16	
G7	251.95	247.23	250.27	250.66	254.22	251.95	251.33	252.66	251.95	
G8	250.82	245.51	249.38	249.53	253.16	251.13	250.39	251.41	250.82	
G9	249.53	244.3	248.2	248.83	252.34	250.35	249.53	250.55	249.53	
G10	248.71	243.09	247.11	246.91	251.56	249.61	247.77	249.22	248.71	
G11	246.72	241.25	245.63	245.55	249.57	248.4	246.99	248.13	246.72	
G12	245.23	240.08	244.3	244.18	248.75	247.81	245.35	246.56	245.23	
G13	243.71	238.13	242.54	242.7	247.34	246.37	243.63	245.16	243.71	
G14	241.95	236.72	241.41	241.29	245.59	244.77	242.42	243.71	241.95	
G15	240.47	234.34	239.8	239.45	244.3	243.24	240.51	242.81	240.47	
G16	238.98	232.07	238.01	237.66	242.62	241.95	239.22	240.78	238.98	
G17	236.48	230.04	236.02	235.59	241.33	240.35	237.11	239.61	236.48	
G18	234.02	227.27	233.98	233.48	239.18	239.02	235.23	237.85	234.02	
G19	231.33	224.73	231.76	230.78	237.15	237.3	233.87	236.05	231.33	
G20	228.83	221.48	228.91	228.44	235.0	235.7	231.17	233.91	228.83	
G21	225.35	218.75	225.98	225.43	232.11	234.26	227.81	231.68	225.35	
G22	222.62	215.2	223.13	222.23	229.38	232.07	225.04	229.22	222.62	
G23	219.49	211.84	220.08	219.53	226.95	230.16	221.88	226.88	219.49	
G24	215.66	208.28	217.23	215.98	223.36	227.27	219.1	223.67	215.66	
G25	212.07	205.04	213.67	212.58	220.51	224.22	215.2	220.35	212.07	
G26	208.32	201.02	210.0	208.4	216.56	221.29	211.48	216.95	208.32	
G27	204.8	196.72	206.33	204.77	213.16	217.93	207.81	213.63	204.8	
G28	200.2	192.73	202.5	200.74	209.06	214.88	204.26	209.88	200.2	
G29	196.09	188.28	198.71	196.72	205.55	211.37	200.16	206.41	196.09	
G30	192.3	183.79	194.69	192.3	201.09	207.54	196.21	202.42	192.3	
G31	187.73	179.26	190.43	187.77	196.64	203.67	192.23	198.63	187.73	
G32	182.89	174.57	185.9	183.36	192.19	199.57	187.23	193.83	182.89	
G33	178.36	169.69	181.25	178.95	187.93	195.74	182.66	189.45	178.36	
G34	173.71	164.84	177.07	173.71	182.93	191.33	178.01	185.08	173.71	
G35	168.98	159.88	171.99	168.79	178.52	187.07	173.36	180.51	168.98	
G36	163.59	154.38	166.91	163.67	173.4	182.23	168.09	175.47	163.59	

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:13:32

LUMINOUS DISTRIBUTION INTENSITY DATA(Continue 1)

Gamma/C	C0	C45	C90	C135	C180	C225	C270	C315	C360	
G37	157.93	149.61	161.84	158.48	168.13	177.77	163.01	171.02	157.93	
G38	152.66	143.75	156.48	153.09	162.73	172.89	157.66	165.51	152.66	
G39	146.84	138.09	151.13	147.7	157.5	167.66	151.99	160.2	146.84	
G40	140.86	132.58	145.63	142.15	152.38	162.38	146.33	154.61	140.86	
G41	135.08	126.84	140.16	136.64	146.68	156.84	140.63	149.22	135.08	
G42	129.41	121.13	134.53	131.02	141.09	151.64	134.77	143.28	129.41	
G43	123.59	115.63	128.55	125.43	135.27	145.66	128.71	137.5	123.59	
G44	117.62	109.41	122.81	119.26	129.8	139.8	123.09	131.56	117.62	
G45	112.07	103.4	117.07	113.28	124.06	134.02	117.19	125.82	112.07	
G46	105.7	97.58	110.94	107.38	118.24	128.01	111.45	119.88	105.7	
G47	100.0	91.91	104.96	101.37	112.66	122.15	105.47	113.91	100.0	
G48	94.18	85.7	99.02	95.55	106.84	116.25	99.8	108.13	94.18	
G49	88.48	79.88	93.05	89.26	101.21	110.31	93.95	102.11	88.48	
G50	83.2	74.14	87.03	83.48	95.31	104.49	88.48	96.29	83.2	
G51	77.77	68.63	81.17	77.7	89.41	98.55	82.89	90.51	77.77	
G52	72.54	63.48	75.35	71.91	83.79	92.85	77.54	84.77	72.54	
G53	67.46	58.2	69.77	66.68	78.32	87.46	72.46	79.3	67.46	
G54	62.58	53.32	64.38	61.45	72.73	81.8	67.23	73.95	62.58	
G55	58.01	48.79	59.1	56.48	67.5	76.45	62.34	68.75	58.01	
G56	53.32	44.65	54.34	51.91	62.42	71.33	57.58	63.83	53.32	
G57	48.98	40.78	49.73	47.58	57.54	66.41	53.09	59.18	48.98	
G58	44.92	37.34	45.55	43.59	52.97	61.6	48.83	54.61	44.92	
G59	40.98	34.18	41.76	39.88	48.67	57.11	44.84	50.47	40.98	
G60	37.54	31.33	38.4	36.52	44.77	52.81	41.37	46.56	37.54	
G61	34.26	28.87	35.23	33.59	41.17	48.87	37.93	42.93	34.26	
G62	31.48	26.72	32.5	31.02	37.85	45.0	35.0	39.65	31.48	
G63	29.06	24.8	30.16	28.83	35.12	41.52	32.42	36.68	29.06	
G64	26.8	23.16	28.09	26.68	32.54	38.36	30.04	34.02	26.8	
G65	24.88	21.76	26.21	24.92	30.35	35.43	28.01	31.45	24.88	
G66	23.28	20.66	24.65	23.36	28.36	32.89	26.09	29.22	23.28	
G67	21.95	19.61	23.32	21.99	26.52	30.59	24.38	27.46	21.95	
G68	20.86	18.67	22.15	20.86	24.88	28.52	22.97	25.55	20.86	
G69	19.96	17.46	21.13	19.88	23.52	26.6	21.68	23.98	19.96	
G70	18.95	16.21	20.0	18.83	22.3	24.96	20.59	22.66	18.95	
G71	17.77	14.88	18.75	17.58	21.33	23.44	19.57	21.56	17.77	
G72	16.41	13.59	17.42	16.17	20.23	22.23	18.67	19.65	16.41	
G73	15.12	12.3	15.98	14.84	19.06	21.17	17.73	19.45	15.12	

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks:

Report No.:

Test Time: 2024/10/15 11:13:32

LUMINOUS DISTRIBUTION INTENSITY DATA(Continue 2)

[illegible]

C Range: 0 - 360 Deg

C Interval: 45.0 Deg

Test Speed: High

Temperature: 6.25

Operators:

Test Date: 2024/10/15 11:13:32

G Range: 0 - 90 Deg

G Interval: 1.0 Deg

Test System: Huipu CPM-2000B Goniophotometer

Humidity: 1

Test Distance: 6.25 m

Remarks: